

SAI HAE NAING LAY

Computer Engineer

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PROFILE

Computer Engineering graduate with hands-on experience in software development and AI and machine learning. Through two internships and a senior project, I have worked on a retail analytics system, a voice-controlled digital signage tool, a LINE ordering system, and a cattle behavior recognition model trained on real farm footage. I work with Python, Node.js, YOLO, OpenCV, and Supabase, and I enjoy learning new technologies and turning ideas into practical, working solutions.

TECHNICAL SKILLS

Programming: C# (Basic), Java, Python

Web / Mobile: HTML, CSS, JavaScript, Node.js, Express, Flutter

AI / ML: YOLO, OpenCV, Mediapipe, Roboflow

Database: MySQL, Supabase, MongoDB

APIs / Tools: REST APIs, LINE Messaging API, LINE LIFF, VS Code, Git

Soft Skills: Teamwork, Time Management, Quick Learner, Adaptability, Collaboration

Languages: Burmese (Native), English (Fluent), Thai (Basic), German (A1)

EXPERIENCE

CAS Cogniser (Thailand) — Chiang Mai, Thailand

Internship Trainee

LINE Bot Sushi Cafe System — Node.js, Express, LINE LIFF, Supabase

Jan – Apr 2026

- Developed a full-stack ordering system with LINE Messaging API integration; implemented role-based dashboards (customer / shop master / admin) with authentication
- Built an order management system with real-time updates and a points-rewards feature

Voice-Activated Digital Signage Control System — Python, Flask

Jun – Jul 2025

- Built a voice-controlled ad display with real-time play / stop / switch functionality and continuous listening with auto-demo fallback
- Created dynamic ad-category mapping with automatic video file detection

Smart Retail Analytics System — YOLO, OpenCV, Face Recognition, NumPy

Jun – Jul 2025

- Built a retail analytics system combining people counting, face recognition, and heat-map generation, with real-time entry / exit tracking in defined zones
- Built a live dashboard with video stream, occupancy metrics, entry / exit logs, and heat-map overlays

PROJECTS

Behavior Recognition for Estrus Monitoring in Cattle

Senior Final Project — Computer Vision, YOLOv8/v11, Python, Roboflow

- Built a computer vision pipeline (dataset construction, Roboflow annotation, YOLO training, evaluation on real farm CCTV footage) to classify cow behaviors and support automated estrus detection

Human Action Recognition System

Team Project — Tkinter, Image Processing, Python, Mediapipe

- Built a human action recognition system using classical image processing — enhancement, segmentation, edge detection, geometric transforms — covering preprocessing through classification

EDUCATION

Mae Fah Luang University

Aug 2022 – July 2026

Bachelor of Engineering in Computer Engineering — Graduation: June 2026 — Status: Fresh Graduate